

Increasing the CPU frequency:

1. Program the new number of wait states to the LATENCY bits in [Section 6.9: FLASH registers](#).
2. Check that the new number of wait states is taken into account to access the Flash memory by reading the FLASH_ACR register.
3. Modify the CPU clock source by writing the SW bits in the RCC_CFGR register.
4. If needed, modify the CPU clock prescaler by writing the HPRE bits in RCC_CFGR.
5. Check that the new CPU clock source or/and the new CPU clock prescaler value is/are taken into account by reading the clock source status (SWS bits) or/and the AHB prescaler value (HPRE bits), respectively, in the RCC_CFGR register.

Decreasing the CPU frequency:

1. Modify the CPU clock source by writing the SW bits in the RCC_CFGR register.
2. If needed, modify the CPU clock prescaler by writing the HPRE bits in RCC_CFGR.
3. Check that the new CPU clock source or/and the new CPU clock prescaler value is/are taken into account by reading the clock source status (SWS bits) or/and the AHB prescaler value (HPRE bits), respectively, in the RCC_CFGR register.
4. Program the new number of wait states to the LATENCY bits in [Section 6.9: FLASH registers](#).
5. Check that the new number of wait states is used to access the Flash memory by reading the FLASH_ACR register.

6.3.4 Low-voltage read

When the external SMPS option is used, the Flash must be programmed in low voltage read mode.

- Voltage scaling range must be in range 2.
- Unlock the LVEN bit by the following procedure:
 - Write LVEKEY1 = 0xF4F5F6F7 in the FLASH_LVEKEYR.
 - Write LVEKEY2 = 0x0A1B2C3D in the FLASH_LVEKEYR.
- Set the LVEN bit in the FLASH_ACR register.
- Check the LVEN bit is set.
- Read back the LVEN bit the FLASH_ACR register

6.3.5 Flash program and erase operations

The embedded Flash memory can be programmed using in-circuit programming or in-application programming.

The **in-circuit programming (ICP)** method is used to update the entire contents of the Flash memory, using the JTAG, SWD protocol or the bootloader to load the user application into the microcontroller. ICP offers quick and efficient design iterations and eliminates unnecessary package handling or socketing of devices.

In contrast to the ICP method, **in-application programming (IAP)** can use any communication interface supported by the microcontroller (I/Os, USB, CAN, UART, I²C, SPI, etc.) to download programming data into memory. IAP allows the user to re-program the